INVESTIGATING THE INFLUENCE OF HOST AGE ON MURINE MICROBIAL RESPONSES TO DIETARY CONTAMINANTS

Amon Cox, Evelyn Callaway, and Arul Jayaraman*

Artie McFerrin Department of Chemical Engineering, Texas A&M University, College Station, Texas, United States of America, 77843

*Correspondence: arulj@tamu.edu

Supplementary Material

Table S1 Dunn's Test for 100 µM BPS Age 22-Weeks Alpha Diversity Scores

Bonferroni-corrected *p* values for each exposure and culture day by alpha-diversity index.

Alpha-Diversity Index	Comparison	Adjusted p value
Shannon	bps_d2 - bps_d7	1.000
Shannon	bps_d2 - control_d2	1.000
Shannon	bps_d7 - control_d2	1.000
Shannon	bps_d2 - control_d7	1.000
Shannon	bps_d7 - control_d7	1.000
Shannon	control_d2 - control_d7	1.000
Shannon	bps_d2 - inoculum_d0	0.155
Shannon	bps_d7 - inoculum_d0	1.000
Shannon	control_d2 - inoculum_d0	0.135
Shannon	control_d7 - inoculum_d0	1.000
Inverse Simpson	bps_d2 - bps_d7	1.000
Inverse Simpson	bps_d2 - control_d2	1.000
Inverse Simpson	bps_d7 - control_d2	1.000
Inverse Simpson	bps_d2 - control_d7	1.000
Inverse Simpson	bps_d7 - control_d7	1.000
Inverse Simpson	control_d2 - control_d7	1.000
Inverse Simpson	bps_d2 - inoculum_d0	0.165
Inverse Simpson	bps_d7 - inoculum_d0	1.000
Inverse Simpson	control_d2 - inoculum_d0	0.115
Inverse Simpson	control_d7 - inoculum_d0	1.000

Table S2 Dunn's Test for 100 µM DINP Age 22-Weeks Alpha Diversity Scores

Bonferroni-corrected *p* values for each exposure and culture day by alpha-diversity index.

Alpha-Diversity Index	Comparison	Adjusted p value
Shannon	control_d2 - control_d7	1.000
Shannon	control_d2 - dinp_d2	1.000
Shannon	control_d7 - dinp_d2	1.000
Shannon	control_d2 - dinp_d7	1.000
Shannon	control_d7 - dinp_d7	1.000
Shannon	dinp_d2 - dinp_d7	1.000
Shannon	control_d2 - inoculum_d0	0.569
Shannon	control_d7 - inoculum_d0	1.000
Shannon	dinp_d2 - inoculum_d0	0.504
Shannon	dinp_d7 - inoculum_d0	1.000
Inverse Simpson	control_d2 - control_d7	1.000
Inverse Simpson	control_d2 - dinp_d2	1.000
Inverse Simpson	control_d7 - dinp_d2	1.000
Inverse Simpson	control_d2 - dinp_d7	1.000
Inverse Simpson	control_d7 - dinp_d7	1.000
Inverse Simpson	dinp_d2 - dinp_d7	1.000
Inverse Simpson	control_d2 - inoculum_d0	0.536
Inverse Simpson	control_d7 - inoculum_d0	1.000
Inverse Simpson	dinp_d2 - inoculum_d0	0.536
Inverse Simpson	dinp_d7 - inoculum_d0	1.000

Table S3 Dunn's Test for 100 µM DINP Age 40-Weeks Alpha Diversity Scores

Bonferroni-corrected *p* values for each exposure and culture day by alpha-diversity index.

Alpha-Diversity Index	Comparison	Adjusted p value
Shannon	control_d2 - control_d7	1.000
Shannon	control_d2 - dinp_d2	1.000
Shannon	control_d7 - dinp_d2	1.000
Shannon	control_d2 - dinp_d7	1.000
Shannon	control_d7 - dinp_d7	1.000
Shannon	dinp_d2 - dinp_d7	1.000
Shannon	control_d2 - inoculum_d0	0.179
Shannon	control_d7 - inoculum_d0	0.069
Shannon	dinp_d2 - inoculum_d0	0.135
Shannon	dinp_d7 - inoculum_d0	0.011
Inverse Simpson	control_d2 - control_d7	1.000
Inverse Simpson	control_d2 - dinp_d2	1.000
Inverse Simpson	control_d7 - dinp_d2	1.000
Inverse Simpson	control_d2 - dinp_d7	1.000
Inverse Simpson	control_d7 - dinp_d7	1.000
Inverse Simpson	dinp_d2 - dinp_d7	1.000
Inverse Simpson	control_d2 - inoculum_d0	0.152
Inverse Simpson	control_d7 - inoculum_d0	0.032
Inverse Simpson	dinp_d2 - inoculum_d0	0.122
Inverse Simpson	dinp_d7 - inoculum_d0	0.012

Table S4 Pairwise Comparisons for 100 µM Exposure Beta-Diversity Scores

Adjusted p values for multilevel pairwise comparisons of Bray-Curtis dissimilarities for 16S microbial abundances. Each comparison is between a 100 µM exposure and culture day across the host ages.

group1	group2	BPS 100µM wk22	DINP 100µM wk22	DINP 100µM wk40
inoculum_d0	control_d2	0.039	0.004	0.273
inoculum_d0	exposure_d2	0.030	0.002	0.190
inoculum_d0	control_d7	0.006	0.002	0.095
inoculum_d0	exposure_d7	0.006	0.003	0.095
control_d2	exposure_d2	0.990	0.972	0.969
control_d2	control_d7	0.005	0.003	0.180
control_d2	exposure_d7	0.007	0.002	0.328
exposure_d2	control_d7	0.005	0.003	0.180
exposure_d2	exposure_d7	0.006	0.002	0.273
control_d7	exposure_d7	0.990	0.972	0.912

Table S5 Age-Dependent, Exposure-Differentiating Pathways

Putative metabolic pathways from LC-MS profiles of microbial cultures (n = 7) that were significantly differentially abundant ($q < .05$, absolute $\log_2$ fold change > 1) by exposure type and across host age (FDR < .05). Each row represents a specific pathway that met these criteria at a given culture day within a dataset (exposure type, dose level, and host age at time of culture inoculation). [1]The fold change of the $\log_2$ pathway intensities (arbitrary units) between exposed and control cultures. [2]A positive slope indicates the pathway increased in baseline intensity (abundance) within the inocula samples as the mice aged, whereas a negative slope indicates the baseline abundance decreased with host age. [3]The number of LC-MS peaks putatively identified as chemical compounds which belong to the given pathway in the given dataset for that comparison.

dataset	culture day	pathwayID	[1]log FC	[2]age trend	[3]features	pathway
BPS_high_wk07	d1	map00760	2.877883	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk07	d2	map00760	2.960512	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk07	d7	map00760	3.106409	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk07	d1	map00907	1.037721	0.0575217	9	Pinene, camphor and geraniol degradation
BPS_high_wk07	d2	map00907	1.002969	0.0575217	9	Pinene, camphor and geraniol degradation
BPS_high_wk07	d2	map04923	1.199016	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk07	d7	map04923	1.399889	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk12	d2	map00522	-1.917188	0.0505457	12	Biosynthesis of 12-, 14- and 16-membered macrolides
BPS_high_wk12	d1	map00760	3.047188	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk12	d2	map00760	3.028977	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk12	d7	map00760	2.984488	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk12	d2	map00903	-1.860440	0.0524729	5	Limonene degradation
BPS_high_wk12	d1	map00945	1.197490	-0.0574412	12	Stilbenoid, diarylheptanoid and gingerol biosynthesis
BPS_high_wk12	d2	map00945	1.966531	-0.0574412	12	Stilbenoid, diarylheptanoid and gingerol biosynthesis
BPS_high_wk12	d7	map00945	1.245410	-0.0574412	12	Stilbenoid, diarylheptanoid and gingerol biosynthesis
BPS_high_wk12	d1	map04923	1.311364	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk12	d2	map04923	1.359272	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk12	d7	map04923	1.106088	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk17	d1	map00760	2.601902	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk17	d2	map00760	2.296052	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk17	d7	map00760	2.572578	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk17	d1	map00945	1.753917	-0.0574412	12	Stilbenoid, diarylheptanoid and gingerol biosynthesis
BPS_high_wk17	d2	map00945	1.488442	-0.0574412	12	Stilbenoid, diarylheptanoid and gingerol biosynthesis
BPS_high_wk17	d1	map00983	1.223532	-0.0465852	6	Drug metabolism - other enzymes
BPS_high_wk17	d2	map00983	1.022960	-0.0465852	6	Drug metabolism - other enzymes
BPS_high_wk17	d1	map01057	1.073861	0.1684322	6	Biosynthesis of type II polyketide products
BPS_high_wk17	d1	map04923	1.044176	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk17	d7	map04923	1.333014	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk22	d7	map00074	1.029566	-0.0327958	12	Mycolic acid biosynthesis
BPS_high_wk22	d2	map00402	1.229830	-0.1464445	15	Benzoxazinoid biosynthesis
BPS_high_wk22	d1	map00621	1.785103	-0.1008060	15	Dioxin degradation
BPS_high_wk22	d2	map00621	1.092638	-0.1008060	15	Dioxin degradation
BPS_high_wk22	d7	map00621	1.422935	-0.1008060	15	Dioxin degradation
BPS_high_wk22	d1	map00760	3.090173	-0.1007148	28	Nicotinate and nicotinamide metabolism
BPS_high_wk22	d2	map00760	2.726602	-0.1007148	28	Nicotinate and nicotinamide metabolism
BPS_high_wk22	d7	map00760	2.994885	-0.1007148	28	Nicotinate and nicotinamide metabolism
BPS_high_wk22	d1	map00901	1.454377	-0.0242287	17	Indole alkaloid biosynthesis
BPS_high_wk22	d7	map00910	1.069787	-0.0968412	9	Nitrogen metabolism
BPS_high_wk22	d1	map00945	2.176640	-0.0574412	13	Stilbenoid, diarylheptanoid and gingerol biosynthesis
BPS_high_wk22	d2	map00945	1.848521	-0.0574412	13	Stilbenoid, diarylheptanoid and gingerol biosynthesis
BPS_high_wk22	d7	map00945	2.005717	-0.0574412	13	Stilbenoid, diarylheptanoid and gingerol biosynthesis
BPS_high_wk22	d1	map04923	1.675338	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk22	d2	map04923	1.773769	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk22	d7	map04923	1.842716	-0.0515491	14	Regulation of lipolysis in adipocytes
BPS_high_wk22	d7	map04927	1.071207	-0.0968268	8	Cortisol synthesis and secretion
BPS_high_wk22	d7	map04934	1.071207	-0.0968268	8	Cushing syndrome
BPS_high_wk22	d2	map04974	1.080131	-0.1422951	30	Protein digestion and absorption
BPS_high_wk40	d1	map00760	1.988032	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk40	d2	map00760	2.161343	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk40	d7	map00760	2.115946	-0.1007148	25	Nicotinate and nicotinamide metabolism
BPS_high_wk40	d7	map00901	-1.191863	-0.0242287	18	Indole alkaloid biosynthesis
BPS_low_wk12	d7	map00450	-1.062807	-0.1114369	3	Selenocompound metabolism
BPS_low_wk12	d7	map00623	-1.062807	-0.0945718	36	Toluene degradation
BPS_low_wk12	d7	map00750	-1.062807	-0.0904201	13	Vitamin B6 metabolism
DINP_high_wk22	d7	map00074	3.521399	0.0405416	12	Mycolic acid biosynthesis
DINP_high_wk22	d7	map00121	3.596630	0.1130709	19	Secondary bile acid biosynthesis
DINP_high_wk22	d7	map00230	1.541767	0.0756323	36	Purine metabolism
DINP_high_wk22	d7	map00290	3.157042	0.2064483	14	Valine, leucine and isoleucine biosynthesis
DINP_high_wk22	d7	map00330	2.266190	0.0730470	38	Arginine and proline metabolism
DINP_high_wk22	d7	map00540	2.247822	0.1374290	3	Lipopolysaccharide biosynthesis
DINP_high_wk22	d7	map00562	1.883385	0.1773897	21	Inositol phosphate metabolism
DINP_high_wk22	d7	map00566	2.040559	-0.1626972	17	Sulfoquinovose metabolism
DINP_high_wk22	d7	map00590	1.918227	0.0112358	56	Arachidonic acid metabolism
DINP_high_wk22	d7	map00620	2.060231	0.0383105	11	Pyruvate metabolism
DINP_high_wk22	d7	map00630	1.259549	0.0744889	28	Glyoxylate and dicarboxylate metabolism
DINP_high_wk22	d7	map00680	2.688622	0.0100724	49	Methane metabolism
DINP_high_wk22	d7	map00720	1.725803	0.0927123	14	Other carbon fixation pathways
DINP_high_wk22	d7	map00730	2.455008	0.0296778	13	Thiamine metabolism
DINP_high_wk22	d7	map00750	1.462533	0.0569476	13	Vitamin B6 metabolism
DINP_high_wk22	d7	map00791	2.134128	-0.1854567	3	Atrazine degradation
DINP_high_wk22	d2	map00901	1.135084	0.0376894	19	Indole alkaloid biosynthesis
DINP_high_wk22	d7	map00901	2.274934	0.0376894	19	Indole alkaloid biosynthesis
DINP_high_wk22	d7	map00904	2.353474	0.0609885	41	Diterpenoid biosynthesis
DINP_high_wk22	d7	map00906	2.223259	0.1806403	20	Carotenoid biosynthesis
DINP_high_wk22	d7	map00920	3.321686	0.1323002	27	Sulfur metabolism
DINP_high_wk22	d7	map00930	2.918177	0.1921522	9	Caprolactam degradation
DINP_high_wk22	d7	map00946	2.894051	0.0704668	42	Degradation of flavonoids
DINP_high_wk22	d7	map00970	1.725803	0.0927123	13	Aminoacyl-tRNA biosynthesis
DINP_high_wk22	d7	map00980	1.765250	0.0317928	35	Metabolism of xenobiotics by cytochrome P450
DINP_high_wk22	d7	map01051	1.786630	0.0066141	16	Biosynthesis of ansamycins

dataset	culture day	pathwayID	[1]log FC	[2]age trend	[3]features	pathway
DINP_high_wk22	d7	map01054	1.995361	0.1452738	5	Nonribosomal peptide structures
DINP_high_wk22	d7	map01063	1.952187	0.0407488	48	Biosynthesis of alkaloids derived from shikimate pathway
DINP_high_wk22	d7	map01100	2.202702	0.0431300	518	Metabolic pathways
DINP_high_wk22	d7	map01110	1.802727	0.0142835	414	Biosynthesis of secondary metabolites
DINP_high_wk22	d7	map01120	2.380088	0.0575691	289	Microbial metabolism in diverse environments
DINP_high_wk22	d7	map01200	2.684454	0.0107213	48	Carbon metabolism
DINP_high_wk22	d7	map01240	2.080708	0.0297197	95	Biosynthesis of cofactors
DINP_high_wk22	d7	map01502	3.545699	0.1327692	17	Vancomycin resistance
DINP_high_wk22	d7	map01523	1.725803	0.0927123	9	Antifolate resistance
DINP_high_wk22	d7	map04024	4.084933	0.0398469	7	cAMP signaling pathway
DINP_high_wk22	d7	map04216	-1.161158	0.0905865	22	Ferroptosis
DINP_high_wk22	d7	map04918	1.725803	0.0927123	9	Thyroid hormone synthesis
DINP_high_wk22	d7	map04921	1.395512	-0.0883922	14	Oxytocin signaling pathway
DINP_high_wk22	d7	map05204	1.567442	0.0312867	17	Chemical carcinogenesis - DNA adducts
DINP_high_wk22	d7	map05208	1.639058	0.0312299	18	Chemical carcinogenesis - reactive oxygen species
DINP_high_wk22	d7	map05310	1.497333	0.0763300	6	Asthma
DINP_high_wk22	d7	map07216	1.995361	0.1452738	3	Catecholamine transferase inhibitors
DINP_high_wk22	d7	map07232	3.124820	0.0894279	3	Potassium channel blocking and opening drugs
DINP_high_wk40	d1	map00040	3.681841	0.0781518	70	Pentose and glucuronate interconversions
DINP_high_wk40	d2	map00040	3.198685	0.0781518	70	Pentose and glucuronate interconversions
DINP_high_wk40	d7	map00040	2.472544	0.0781518	70	Pentose and glucuronate interconversions
DINP_high_wk40	d1	map00052	6.710070	0.0272990	27	Galactose metabolism
DINP_high_wk40	d2	map00052	5.392945	0.0272990	27	Galactose metabolism
DINP_high_wk40	d7	map00052	1.819439	0.0272990	27	Galactose metabolism
DINP_high_wk40	d1	map00053	3.681841	0.0764881	54	Ascorbate and aldarate metabolism
DINP_high_wk40	d2	map00053	3.198685	0.0764881	54	Ascorbate and aldarate metabolism
DINP_high_wk40	d7	map00053	2.472544	0.0764881	54	Ascorbate and aldarate metabolism
DINP_high_wk40	d1	map00074	2.803729	0.0405416	14	Mycolic acid biosynthesis
DINP_high_wk40	d2	map00074	2.645810	0.0405416	14	Mycolic acid biosynthesis
DINP_high_wk40	d7	map00074	2.680788	0.0405416	14	Mycolic acid biosynthesis
DINP_high_wk40	d1	map00121	9.305250	0.1130709	23	Secondary bile acid biosynthesis
DINP_high_wk40	d2	map00121	8.498944	0.1130709	23	Secondary bile acid biosynthesis
DINP_high_wk40	d7	map00121	6.622731	0.1130709	23	Secondary bile acid biosynthesis
DINP_high_wk40	d1	map00240	3.681841	0.0502201	55	Pyrimidine metabolism
DINP_high_wk40	d2	map00240	3.198685	0.0502201	55	Pyrimidine metabolism
DINP_high_wk40	d7	map00240	2.472544	0.0502201	55	Pyrimidine metabolism
DINP_high_wk40	d1	map00290	3.357487	0.2064483	16	Valine, leucine and isoleucine biosynthesis
DINP_high_wk40	d2	map00290	2.270012	0.2064483	16	Valine, leucine and isoleucine biosynthesis
DINP_high_wk40	d7	map00290	2.525097	0.2064483	16	Valine, leucine and isoleucine biosynthesis
DINP_high_wk40	d1	map00330	3.514973	0.0730470	42	Arginine and proline metabolism
DINP_high_wk40	d2	map00330	2.606762	0.0730470	42	Arginine and proline metabolism
DINP_high_wk40	d7	map00330	2.637019	0.0730470	42	Arginine and proline metabolism
DINP_high_wk40	d1	map00410	3.681841	0.0187635	27	beta-Alanine metabolism
DINP_high_wk40	d2	map00410	3.198685	0.0187635	27	beta-Alanine metabolism
DINP_high_wk40	d7	map00410	2.472544	0.0187635	27	beta-Alanine metabolism
DINP_high_wk40	d2	map00540	1.386533	0.1374290	5	Lipopolysaccharide biosynthesis
DINP_high_wk40	d7	map00540	1.533255	0.1374290	5	Lipopolysaccharide biosynthesis
DINP_high_wk40	d1	map00562	4.979104	0.1773897	23	Inositol phosphate metabolism
DINP_high_wk40	d2	map00562	4.804989	0.1773897	23	Inositol phosphate metabolism
DINP_high_wk40	d7	map00562	3.281044	0.1773897	23	Inositol phosphate metabolism
DINP_high_wk40	d1	map00572	6.710070	0.0562205	5	Arabinogalactan biosynthesis - Mycobacterium
DINP_high_wk40	d2	map00572	5.392945	0.0562205	5	Arabinogalactan biosynthesis - Mycobacterium
DINP_high_wk40	d7	map00572	1.819439	0.0562205	5	Arabinogalactan biosynthesis - Mycobacterium
DINP_high_wk40	d1	map00590	3.086814	0.0112358	64	Arachidonic acid metabolism
DINP_high_wk40	d2	map00590	2.803310	0.0112358	64	Arachidonic acid metabolism
DINP_high_wk40	d7	map00590	2.464942	0.0112358	64	Arachidonic acid metabolism
DINP_high_wk40	d1	map00620	4.046117	0.0383105	13	Pyruvate metabolism
DINP_high_wk40	d2	map00620	4.084053	0.0383105	13	Pyruvate metabolism
DINP_high_wk40	d7	map00620	2.895051	0.0383105	13	Pyruvate metabolism
DINP_high_wk40	d2	map00623	1.370397	0.0891112	38	Toluene degradation
DINP_high_wk40	d7	map00623	1.519811	0.0891112	38	Toluene degradation
DINP_high_wk40	d1	map00680	3.513015	0.0100724	53	Methane metabolism
DINP_high_wk40	d2	map00680	3.364435	0.0100724	53	Methane metabolism
DINP_high_wk40	d7	map00680	2.724059	0.0100724	53	Methane metabolism
DINP_high_wk40	d1	map00730	2.803729	0.0296778	15	Thiamine metabolism
DINP_high_wk40	d2	map00730	2.645810	0.0296778	15	Thiamine metabolism
DINP_high_wk40	d7	map00730	2.680788	0.0296778	15	Thiamine metabolism
DINP_high_wk40	d1	map00901	1.155368	0.0376894	20	Indole alkaloid biosynthesis
DINP_high_wk40	d2	map00901	1.033094	0.0376894	20	Indole alkaloid biosynthesis
DINP_high_wk40	d1	map00904	4.633131	0.0609885	47	Diterpenoid biosynthesis
DINP_high_wk40	d2	map00904	4.310574	0.0609885	47	Diterpenoid biosynthesis
DINP_high_wk40	d7	map00904	3.775822	0.0609885	47	Diterpenoid biosynthesis
DINP_high_wk40	d1	map00906	4.979104	0.1806403	22	Carotenoid biosynthesis
DINP_high_wk40	d2	map00906	4.804989	0.1806403	22	Carotenoid biosynthesis
DINP_high_wk40	d7	map00906	3.281044	0.1806403	22	Carotenoid biosynthesis
DINP_high_wk40	d1	map00920	9.305250	0.1323002	29	Sulfur metabolism
DINP_high_wk40	d2	map00920	8.498944	0.1323002	29	Sulfur metabolism
DINP_high_wk40	d7	map00920	6.622731	0.1323002	29	Sulfur metabolism
DINP_high_wk40	d1	map00930	3.357487	0.1921522	11	Caprolactam degradation
DINP_high_wk40	d2	map00930	2.270012	0.1921522	11	Caprolactam degradation
DINP_high_wk40	d7	map00930	2.525097	0.1921522	11	Caprolactam degradation
DINP_high_wk40	d1	map00946	3.698429	0.0704668	46	Degradation of flavonoids
DINP_high_wk40	d2	map00946	2.852644	0.0704668	46	Degradation of flavonoids
DINP_high_wk40	d7	map00946	2.373815	0.0704668	46	Degradation of flavonoids
DINP_high_wk40	d1	map00980	5.153258	0.0317928	37	Metabolism of xenobiotics by cytochrome P450
DINP_high_wk40	d2	map00980	6.078218	0.0317928	37	Metabolism of xenobiotics by cytochrome P450
DINP_high_wk40	d7	map00980	4.557054	0.0317928	37	Metabolism of xenobiotics by cytochrome P450
DINP_high_wk40	d1	map00998	1.076584	-0.0059666	45	Biosynthesis of various antibiotics
DINP_high_wk40	d2	map01051	1.028114	0.0066141	17	Biosynthesis of ansamycins
DINP_high_wk40	d1	map01061	1.220613	-0.0360118	64	Biosynthesis of phenylpropanoids
DINP_high_wk40	d2	map01061	1.083139	-0.0360118	64	Biosynthesis of phenylpropanoids
DINP_high_wk40	d1	map01100	3.852846	0.0431300	570	Metabolic pathways
DINP_high_wk40	d2	map01100	3.640761	0.0431300	570	Metabolic pathways
DINP_high_wk40	d7	map01100	2.928177	0.0431300	570	Metabolic pathways
DINP_high_wk40	d1	map01110	2.524306	0.0142835	447	Biosynthesis of secondary metabolites
DINP_high_wk40	d2	map01110	2.397483	0.0142835	447	Biosynthesis of secondary metabolites
DINP_high_wk40	d7	map01110	1.887231	0.0142835	447	Biosynthesis of secondary metabolites
DINP_high_wk40	d1	map01120	4.671624	0.0575691	322	Microbial metabolism in diverse environments
DINP_high_wk40	d2	map01120	4.398841	0.0575691	322	Microbial metabolism in diverse environments
DINP_high_wk40	d7	map01120	3.710773	0.0575691	322	Microbial metabolism in diverse environments
DINP_high_wk40	d1	map01200	3.944421	0.0107213	50	Carbon metabolism

dataset	culture day	pathwayID	[1]log FC	[2]age trend	[3]features	pathway
DINP_high_wk40	d2	map01200	3.750832	0.0107213	50	Carbon metabolism
DINP_high_wk40	d7	map01200	1.871726	0.0107213	50	Carbon metabolism
DINP_high_wk40	d1	map01240	5.530867	0.0297197	103	Biosynthesis of cofactors
DINP_high_wk40	d2	map01240	4.830807	0.0297197	103	Biosynthesis of cofactors
DINP_high_wk40	d7	map01240	4.469599	0.0297197	103	Biosynthesis of cofactors
DINP_high_wk40	d1	map01502	9.305250	0.1327692	19	Vancomycin resistance
DINP_high_wk40	d2	map01502	8.498944	0.1327692	19	Vancomycin resistance
DINP_high_wk40	d7	map01502	6.622731	0.1327692	19	Vancomycin resistance
DINP_high_wk40	d1	map04024	2.803729	0.0398469	9	cAMP signaling pathway
DINP_high_wk40	d2	map04024	2.645810	0.0398469	9	cAMP signaling pathway
DINP_high_wk40	d7	map04024	2.680788	0.0398469	9	cAMP signaling pathway
DINP_high_wk40	d1	map04921	3.681841	-0.0883922	16	Oxytocin signaling pathway
DINP_high_wk40	d2	map04921	3.198685	-0.0883922	16	Oxytocin signaling pathway
DINP_high_wk40	d7	map04921	2.472544	-0.0883922	16	Oxytocin signaling pathway
DINP_high_wk40	d1	map05204	5.212340	0.0312867	21	Chemical carcinogenesis - DNA adducts
DINP_high_wk40	d2	map05204	5.886731	0.0312867	21	Chemical carcinogenesis - DNA adducts
DINP_high_wk40	d7	map05204	4.518187	0.0312867	21	Chemical carcinogenesis - DNA adducts
DINP_high_wk40	d1	map05208	4.876027	0.0312299	20	Chemical carcinogenesis - reactive oxygen species
DINP_high_wk40	d2	map05208	5.901640	0.0312299	20	Chemical carcinogenesis - reactive oxygen species
DINP_high_wk40	d7	map05208	4.574349	0.0312299	20	Chemical carcinogenesis - reactive oxygen species